HW 2

September 10, 2024

1. Determine whether the sequence converges or diverges. If it converges, find the limit.

$$(1)a_n = \frac{3+5n^2}{n+n^2} \quad (2)a_n = \frac{3+5n^2}{1+n} \quad (3)a_n = \frac{n^4}{n^3-2n}$$
$$(4)a_n = \sqrt{\frac{1+4n^2}{1+n^2}} \quad (5)a_n = \frac{n^2}{\sqrt{n^3+4n}}$$

2. Determine whether the sequence converges or diverges. If it converges, find the limit.

$$(1)a_n = 2+(0.86)^n \quad (2)a_n = 3^n 7^{-n} \quad (3)a_n = \frac{4^n}{1+9^n}$$
$$(4)a_n = \cos\left(\frac{n\pi}{n+1}\right) \quad (5)\{\sin n\}$$

3. Determine whether the sequence converges or diverges. If it converges, find the limit.

$$(1)a_n = \ln(n+1)-\ln n \quad (2)a_n = n\sin(1/n) \quad (3)a_n = \left(1+\frac{2}{n}\right)^n$$
$$(4)a_n = \ln\left(2n^2+1\right)-\ln\left(n^2+1\right) \quad (5)a_n = n-\sqrt{n+1}\sqrt{n+3}$$

4. (a) Determine whether the sequence defined as follows is convergent or divergent:

$$a_1 = 1 \quad a_{n+1} = 4-a_n \quad \text{for } n \geq 1$$

(b) What happens if the first term is $a_1 = 2$?

5. Let $\{a_n\}$ and $\{b_n\}$ be sequences. Assume that $\lim\limits_{n\to\infty} a_n = L$ and $\lim\limits_{n\to\infty} b_n = \mathrm{M}$. Prove:
(a) $\lim_{n\to\infty} a_n b_n = LM$.
(b) $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{L}{M}$ if $M \neq 0$.

6. For what values of r is the sequence $\{nr^n\}$ convergent?

7. (a) If $\{a_n\}$ is convergent, show that

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} a_n$$

(b) A sequence $\{a_n\}$ is defined by $a_1 = 1$ and $a_{n+1} = 1/(1+a_n)$ for $n \geqslant 1$. Assuming that $\{a_n\}$ is convergent, find its limit.

8. Find the limit of the sequence

$$\{\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \ldots\}$$

9.Show that the sequence defined by

$$a_1 = 1 \quad a_{n+1} = 3 - \frac{1}{a_n}$$

is increasing and $a_n < 3$ for all n. Deduce that $\{a_n\}$ is convergent and find its limit.

Bonus

1. Prove that if $\{x_n\}$ is a Cauchy sequence, so is $\{|x_n|\}$.
2. Prove that

$$x_n = a_0 + a_1 q + a_2 q^2 + \cdots + a_n q^n$$

is a Cauchy sequence, where $|q| < 1$ and $|a_k| \leq M, k = 1, 2, \ldots$, for some constant M.